

Jake Golden

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Education

Duke University – Pratt School of Engineering | Durham, NC

Aug 2023 - Present

- B.S.E. in Biomedical Engineering & Electrical and Computer Engineering, expected May 2027
- Cumulative GPA: 4.00/4.00

Experience

Analog Design Intern | National Instruments (Emerson), Austin, TX

May 2026 - Aug 2026

- Designed schematic and layout for precision LCR/SMU characterization PCB.
- Implemented low-leakage switching architecture in guarded multiplexed signal chain.
- Authored 15+ V&V tests for LCR/SMU meter, automated instrumentation control and data acquisition/processing using LabVIEW, compiled results in report.

Product Development Intern | Stryker, Mahwah, NJ

May 2025 - Aug 2025

- Iteratively developed 9 prototypes for novel knee replacement implant: created design inputs to address unmet user needs and modeled in PTC Creo.
- Verified geometric tolerancing and dimensioning equivalency to legacy devices.
- Automated surgical template generation (MATLAB/Creo), saving 10+ hours of work.

Computational Neuroscience Researcher | Brain Stimulation Engineering Lab at Duke University Medical Center, Durham, NC

Jan 2024 - Present

- Program finite element analysis (FEA) ECT/TMS brain stimulation models in MATLAB
- Develop automated pipelines (MATLAB/Bash) that integrate MRI processing, mesh generation/processing (4M+ elements), and coil optimization for high-throughput, patient-specific FEA simulations (~50 simulations/week).

Mechanical Engineering Intern | SIMS Pump Valve Company, Hoboken, NJ

Sep 2022 - Jun 2023

- Created ISO-standard CAD models/drawings using surface modeling in SOLIDWORKS.
- Built standardized part library (300+ parts) for efficient modeling in future projects.
- Performed tolerance verification on pump parts using calipers and micrometers.

Personal Projects

Visit <https://jake-golden.github.io> for more

MATLAB Pulse Oximeter

- Designed single-supply analog front-end conditioning circuit (TIA + active filtering + biased gain) converting photodiode current to PPG voltage waveform
- Implemented custom bidirectional Arduino-MATLAB serial protocol with fixed-length packets, start-byte framing, and checksum validation
- Developed MATLAB DSP (filtering, amplitude extraction, R-ratio-to-SpO₂ calibration) and GUI layer with live plotting, LED control, and diagnostics console

PID-Controlled Self-Balancing Robot

- Built two-wheeled self-balancing robot using Arduino, IMU sensors, and DC motors
- Implemented I²C communication with custom PCB and housing.
- Iteratively tuned PID control parameters to achieve stable balancing.

Skills

Analog/Digital Circuit Design

SPICE simulation (LTSpice)

Op-amp circuit design, filter design, feedback, stability analysis

PCB Design: Schematic capture and layout (KiCad, Xpedition)

Basic CPU design, combinational/sequential Boolean logic

Power electronics & power supply design

In progress: Analog IC design and layout (Cadence Virtuoso)

Mechanical Design

CAD: SOLIDWORKS, PTC Creo, Autodesk Fusion

Fabrication: 3D printing (FDM), laser cutting

Programming

MATLAB (simulation, automation, FEA)

Python, C, Java, LabVIEW

Applied Mathematics

Signal processing:

Time/frequency-domain analysis, filters, control systems

Wave systems: Fourier analysis, transmission line theory, electromagnetic waves

Experimental design:

Hypothesis testing, inferential statistics

Awards & Honors

Pratt School of Engineering
Dean's List with Distinction (2024, 2025, 2026)

Activities & Interests

Duke University football team equipment manager

Part-time math tutor

Piano, guitar

Photography

College basketball enthusiast